

FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	Sb-0000000035
Award Program	Smart Broiler
Project Title	Development of an automated broiler gait score and flock activity assessment system
Grant Start Date	04/01/2020
Grant End Date	03/31/2022
Total Grant Budget	\$ 513,214.32
Total FFAR Budget	\$ 350,000
Matching Funder(s)	Peco Foods, Inc.
Final Report Date	05/01/2022
Project Director/Principal Investigator Information	
Full Name	Hao Gan
Email	Hgan1@utk.edu
Phone	8659748379
Grantee Organization Information	
Organization Name	University of Tennessee
Address	225 Morgan Hall, 2621 Morgan Circle Dr.
City, State Zip	Knoxville, TN 37996



Tax ID	62-6001636
Authorized Signing Official (ASO) Information	
Name	Hollie Renee Schreiber
ASO Title	Director
Email	aggrant@utk.edu
Phone	8659747375

General Information

- 1.1. Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the US, please list the city and country.

Knoxville, Tennessee, Congressional district: TN-002

- 1.2. How many new jobs were created because of this grant funding?

None

- 1.3. How many jobs were maintained because of this grant funding?

None

- 1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain.

No

- 1.5. Has your organization undergone a recent name change? If so, please provide the new name of your organization.

No

2. Scientific Report

- 2.1. Public Abstract (up to 500 words): The abstract should be 'stand-alone' and is

intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.

Consumers worldwide are justifiably concerned with food animal welfare and seek to make purchasing decisions that result in improved rearing practices. A challenge in addressing this concern is the need to develop tools that quantify food animal behavior, the ultimate welfare indicator, rather than the simple environmental welfare indicators currently used, such as the living space afforded an animal.

Our main objective was to develop proof-of-concept vision systems to assess (1) gait scores of individual broilers and (2) gait scores of the broiler flocks based on real-time flock activity and distribution indices. Our design approach incorporated vision software training with input from professional welfare assessors to produce an assessment system that solves the expense and labor shortages associated with current manual welfare assessments but maintains the accuracy and integrity of manual assessments. Such an automated system works continuously with high reduction of labor hours for welfare assessment and reducing risk for disease transfer by humans. The system output was flock benchmark score for use by the farmer, food retailer, and ultimately the consumer. The impact of system development resides mainly in providing farmers, for the first time, with a tool to improve rearing practices that result in better animal welfare conditions. As well, the impact extends to the retailer and ultimately the consumer by providing a meaningful animal-based metric to help purchasing decisions that support improved rearing practices.

- 2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-wordlimit).

A side-view system that mimics human-based gait score assessments was



designed to monitor the broilers for 12 – 16 hours a day. A total of 200 broilers per day were randomly sampled by each camera. Broilers' body gestures during walking were tracked and machine learning algorithms were developed to predict the gait scores of each sampled broiler. A daily summary of broilers' gait scores including the gait score distribution and average gait score will be reported. The results achieved by the side-view system can be applied in any commercial broiler farm to monitor sampled sections in the farm and provide an assessment of gait scores of a large number of broilers daily. The information will allow farmers to evaluate the leg health conditions and the temporal changes in broilers leg health. The information will be critical to help farmers to identify potential problems.

The instantaneous activity and distribution indices of the broilers in predefined regions were calculated based on the bird-representing pixel distributions and changes in two consecutive images captured by the top-view system. Regression models were developed using daily mean activity and distribution indices, bird age, and bird body weight. The developed models will allow users to estimate the daily average gait score with outputs of the stationary vision-based system and broiler production data. This system provide an overview of the broilers activity in the farm and an overall estimation of the daily gait score of broilers.

Both the side-view and top-view systems provide gait score estimations of broilers in commercial farms, and the combination of them will allow farmers to get detailed information in terms of broiler leg health. The information can be used to improve farm management.

- 2.3. Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is to answer questions such as "How did this project lead to improvements in U.S. agriculture and food systems?" or "How can the findings of this study guide or make improvement in future investigations and research?" Outcomes should be explained and classified in one of the following ways:
- a) potential outcomes, i.e., findings, results, or recommendations that could impact U.S. agriculture and food systems, if used;
 - b) b. intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and
 - c) c. end outcomes, i.e., how findings, results, or recommendations

have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.

In this project, two computer vision systems were developed to provide an automated assessment of broiler gait scores in commercial farms. The systems are low cost, requires minimum maintenance, and were designed to be used for most commercial broiler farms. The potential impact of the research is to provide farmers with an automated tool for accurate and timely broiler welfare evaluation. It will lead to improvements in farm management, thus, improvements in animal welfare and health. It will eventually help improve animal productions and will also bring in economic benefits to US agriculture and food systems.

- 2.4. What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit)

The goals of this project include (1) develop a side-view vision system for individual broiler gait score assessments, and (2) develop a top-view vision system for flock level gait score estimation.

Due to the impact of COVID-19, the project was delayed. The changes in the timeline are shown in the table below.

Activities, objectives and Goals	Previous Timeline	Final Timeline
Design and install the close-range vision-based system at UT	04/20--10/20	04/20--12/20
Design and install the ceiling-mounted vision-based system at MSU	04/20--10/20	04/20--12/20
Test the close-range vision-based system and develop statistical models for gait score predictions	07/20--04/21	01/21--09/21
Test the ceiling-mounted vision-based system and develop statistical models for flock activity and gait score predictions	07/20--04/21	01/21--09/21

Integrate the close-range vision-based system and the ceiling-mounted vision-based system at UT.	05/21--09/21	10/21--03/22
Test the integrated system, finalize the system and the models, and present evaluations on the performance of the system.	05/21--09/21	10/21--03/22

- 2.5. Have any of the major goals/specific aims or milestones for the project changed since the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit)

There is no change in project objectives. The team developed and provided the lab-based evaluations of the side-view and top-view systems. Both systems were installed and tested in the same room for monitoring the broilers.

- 2.6. What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit)

The vision systems developed by our team assess the gait scores of broilers at both individual-bird level and flock level.

A video processing pipeline that includes three steps and two light-weighted deep learning models were developed for the side-view system. Firstly, a detection algorithm monitors every broiler in the field of view of the camera and selects the broilers that walked for a duration of more than two seconds.

Secondly, a key body-parts tracking model detects the head, tail, body center, upper legs and feet of the selected broiler, and tracks the coordinates of the key body-parts. Thirdly, a time-series model analyzes the changes of the key body-



parts within the 2-second video (60 video frames) and assign a gait score to the broiler. The system achieved very high accuracy in both models: 2-pixel accuracy in key body-parts tracking (in a 1920×1080 video frame), and 95% in gait score classification. The throughput of the system depends on the configuration of the computer. The team designed the algorithm with the goal of minimizing the needed processing power, and utilized light-weighted deep learning models. It took 6 seconds to process each broiler using a Dell computer (Dell Precision 7820, 16GB GPU). The processing time will decrease significantly if more power GPUs in AWS are used. However, with the goal of sampling 800 broilers each day (from four cameras), there is no need to utilize more computing power.

Our top-view camera system and algorithms were able to monitor broiler activities, distributions, and gait scores across the entire production cycles. This system was designed to realize flock-level monitoring in this phase but is capable to be expanded for individual-bird monitoring with future supports. The accuracies of detecting broiler activities were 99.2% in feeder area and 98.1% in drinking area. In open litter area, the accuracy of activity monitoring varied depending on the video frame rate applied (100% at 25 fps and 84% at 5 fps). Multiple gait score prediction models were developed to accommodate grow-out farms with or without auto-scale for body-weight measurement. The coefficient of determination (R^2) of these models were 0.90 – 0.93 based on our lab-scale trials. The top-view camera system has been installed at the Tyson's research farm at Springdale, AR. With 12 cameras installed at 10' above the floor, we are able to monitor total floor areas of 4,800 ft² which translates into approximately 5,400 broilers (28% of the entire flock). Instead of wide-angle cameras, we used cameras with 90-degree angle of view for our system. The reason of using these regular-angle cameras is because they can produce more accurate estimation on broiler activity and distribution.

- 2.7. Discuss efforts taken to ensure the approach is scientifically rigorous. Include approaches taken to ensure robust and unbiased results.

To ensure the approach is scientifically rigorous, the team raised four flocks with a total of 850 broilers, and set up the lab space to simulate commercial farm conditions. The team also created training and validation datasets when training machine learning models to ensure the models were not biased or



overfitting.

For the side-view vision system, two commercial broiler strains, Cobb500 and Ross708 were used. The floor of the lab was covered by rubber mats to prevent slipping. Then, a layer of commercial broiler litter was applied. Broiler lighting, drinker and feeder were all set up to simulate the condition in commercial broiler farms. The broilers were raised from January 19th to March 26th, 2021, and from May 19th to July 26th, and a camera system was setup to record the broilers from the beginning to the end. The camera was set up at a height of approximately 1.3m with an angle of 30 degrees relative to the ground to mimic the view of a human observer during manual gait scoring in broiler farms. The area that the camera was monitoring was an 8' × 16' pen with 129 broilers. Two identical pens were built for Cobb500 and Ross708. In total, 258 broilers, 129 Cobb500 and 129 Ross708, were used in this study.

For the top-view vision system, Cobb500 and Ross708 were also used. The floor of the lab was covered by rubber mats to prevent slipping. Then, a layer of commercial broiler litter was applied. Broiler lighting, drinker and feeder were all set up to simulate the condition in commercial broiler farms. The broilers were raised from January 19th to March 26th, 2021, and from May 19th to July 26th. A total of 16 pens of size 1.5m × 1.5 m with holding capacity of 12 broilers/pen were built.

- 2.8. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write "nothing to report." A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products. (up to 1,000-word limit)



The results of this project were disseminated through journal publications and conference presentations. Publications and presentations are listed in 3.3.

- 2.9. Describe challenges or delays encountered during project and actions that were taken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)

The team encountered a roughly 6-month delay due to the impact of the COVID-19. Laboratory experiments were forced to stop in 2020 so the team could not raise broilers as we originally planned. In addition, a recruited postdoctoral research associate was not able to come to the U.S. because of travel bans. The team requested a 6-month no cost extension and shifted the timeline (approved) to finish the project in 2022.

- 2.10. What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state "Nothing to Report." For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to 500-word limit)

Nothing to Report.

- 2.11. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 500-word limit)

1 postdoctoral scholar: worked on developing deep learning algorithms to assess individual broiler gait scores.

2 Ph.D. students: work on the development of the side-view vision system and the top-view vision system, as well as the flock level gait score prediction models.

1 research associate: provided daily care and weekly assessment of the broilers



in our animal experiment lab.

3 undergraduate students: provide daily care weekly assessment of the broilers in our animal experiment lab.

3. Information Products

- 3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award.

Information products that have been published include journal papers and conference proceedings.

- 3.2. Please provide a list of citations for the information products produced during the project.

- Yang, X., Y. Zhao, G.M. Street, Y. Huang, S.D.F. To, J.L. Purswell. 2021. Classification of broiler behaviors using triaxial accelerometer and machine learning. *Animal*, 15(7), 100269.
- Yang, X., Y. Zhao, G.T. Tabler. 2020. Accuracy of broiler activity index as affected by sampling time interval. *Animals*, 10(6), 1102.
- Yang, X., Y. Zhao, H. Gan, S. Hawkins, L. Eckelkamp, M. Prado, R. Burns. 2021. Broiler gait score in relation to bird body weight, activity, and age. 2021 International Symposium on Animal Environment and Welfare, Rongchang, Chongqing, China.
- Nasiri, A., Yoder, J., Zhao, Y., Hawkins, S., Prado, M., & Gan, H. (2022). Pose estimation-based lameness recognition in broiler using CNN-LSTM network. *Computers and Electronics in Agriculture*, 197, 106931.

- 3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the project resulting directly from the FFAR award? If yes, please provide citation.



Publications produced from the FFAR award:

- Yang, X., Y. Zhao, G.M. Street, Y. Huang, S.D.F. To, J.L. Purswell. 2021. Classification of broiler behaviors using triaxial accelerometer and machine learning. *Animal*, 15(7), 100269.
- Yang, X., Y. Zhao, G.T. Tabler. 2020. Accuracy of broiler activity index as affected by sampling time interval. *Animals*, 10(6), 1102.
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- 3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.

None.

- 3.5. Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s).

None

- 3.6. Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit)

Since the Phase I period of this project is to develop a proof-of-concept system, no products were confidential, proprietary, or subject to special license



agreements. In the future, the system will be tested in commercial farms and video data collected from those farms will be kept confidential or subject to agreements.

- 3.7. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that other can access and re-use these data in the future?

Currently, information products are disseminated through journal and conference publications. In the future, we plan to publish the model trained and validated in commercial farms through scientific publications and use USDA database Ag Data Commons, to publish broiler datasets related gait scoring.

4. Data Management

- 4.1. Did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imagery). Please list the data generated for the award.

The project generated (unpublished) image/video datasets for training deep learning models and regression models.

- 4.2. If you list multiple data sets, are these data sets related? If so, please provide a short description of how they are related.

The datasets were generated from the same lab experiments. Two datasets were generated from the side-view vision system and the top-view vision system, respectively. They were generated under the same experiment, which can be used to compare the two systems and validate the accuracy of the results from the two systems.

- 4.3. Please provide copies of relevant metadata records to support FFAR's mission



of enhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if necessary, in a compressed folder.

The datasets have not been published. Currently, the datasets are saved in local storage devices and cloud storage services (e.g., Google Drive).

In training the deep learning model for individual broiler gait score assessment, a dataset that contains 9,214 frames was created. Key body parts of broilers in each frame were labeled. This dataset can be used to train different models to estimate broiler gait scores. The dataset set will be expanded to include more data from commercial farms. We plan to share the dataset, eventually.



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RESEARCH

Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name: _____ Hao Gan _____

PI Signature: _____  _____ Date: _____ 6/26/2022 _____

Authorized Signing Official (ASO) Name: Hollie R. Schreiber

ASO Signature: _____  _____ Date: 6/29/22